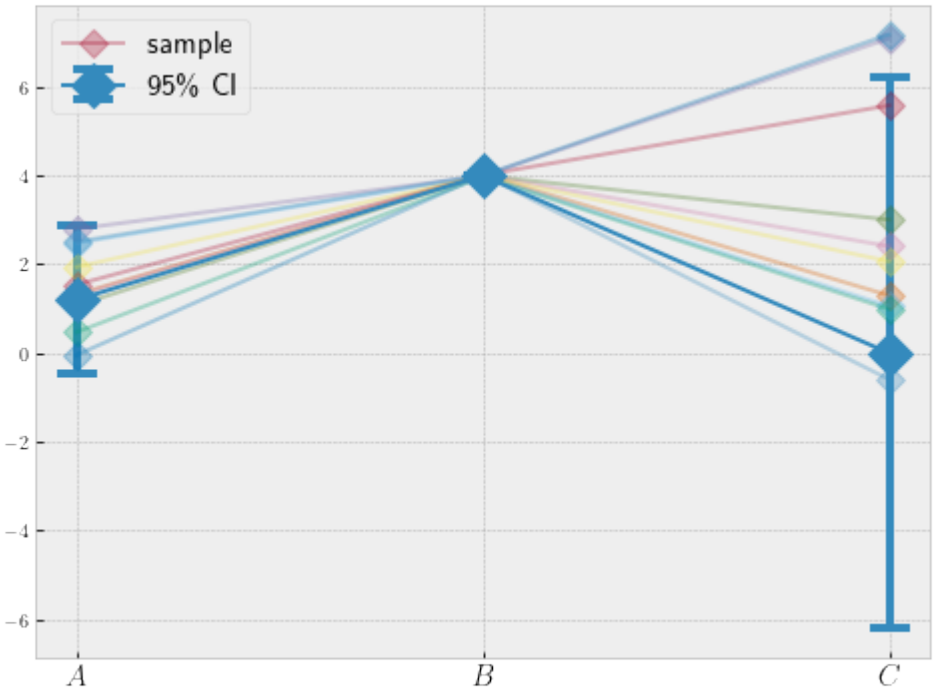


Figure 2.6 Parallel coordinates plot visualizing the multivariate Gaussian from Figure 2.5, conditioned on B = 4. Here, the distribution of A is updated, and all drawn samples interpolate B = 4.

The result is shown in Figure 2.6. Here, upon updating our belief with an observation about B, a few things have changed in our posterior belief:

- The distribution of A changes, taking on a slightly larger mean value due to the positive correlation between A and B. Further, its error bars now have a smaller range.
- The posterior distribution of B simply becomes a special normal distribution with zero variance, since we know for sure what its value is in the posterior. In other words, there is no uncertainty about the value of B anymore.
- Meanwhile, the distribution of C stays the same after the update, as it has no correlation with B.

All of this makes sense, and corresponds to our intuition from the housing price example. Specifically, when we find out about our neighbor's house, the belief about our own house is updated to be similar to the observed price, and our uncertainty also decreases.

What happens when we condition on C then? As you might have guessed, for the same reason that C stays the same after conditioning on a value of B, both the posterior distribution of A and that of B remain unchanged when we condition on C. Figure 2.7 shows this for C = 4.

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